

We are making a remote controlled sailboat using the new vase mode + ribs that we developed. Here is a brief explanation of what is new compared to traditional vase mode with ribs and our, and a tutorial. Make sure to watch the video here: <https://drive.google.com/drive/folders/1verP1uFe15y1qu4991YVATYrRwpX1hgG> it explains the design process with visuals.

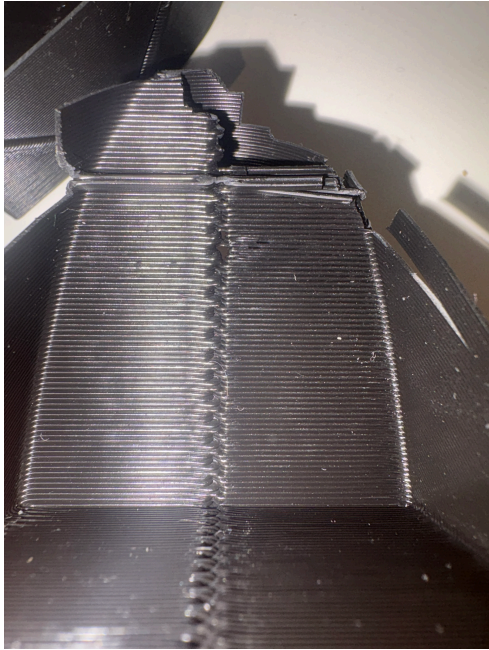
Traditional vase mode with ribs parts have a vertical planar divide that splits the ribs in half, this severely weakens parts made like this, in bending and shear stress. The two main reasons for using vase mode and ribs are to produce lightweight parts and to enable the use of foaming filaments that are hard to print not in vase mode. This means they are often used for wings and fuselages of 3d printed remote controlled planes, where the wings are subject to bending loads. As such the divide is a limiting factor on remote control planes. We are making a remote control sailboat which similarly experiences large bending moments.



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To fix this issue we have developed a system wherein the plane of the divide alternates by a few millimetres every 2 layers, as such a zipper seam is created over where the divide would usually be, largely eliminating the weakness.



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